

Week 06 — Fourier Theory

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The big picture. The Fourier transform (continuous, discrete, finite-data) and the convolution theorem move us from time to frequency; beware aliasing.

The Fourier transform is the bridge between the **time domain** (a function or a sequence g_t) and the **frequency domain** (its transform $G(f)$). We build it in three stages: continuous time (the whole function), discrete time (a sampled sequence), then **finite data** (what we actually observe). Throughout we use ordinary frequency f (not angular frequency $\omega = 2\pi f$). This machinery is the toolbox we will reuse next week to define the spectral density.

Key definitions

Definition — Continuous-time Fourier transform

For $g : \mathbb{R} \rightarrow \mathbb{C}$ with $g \in L^1$,

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i f t} dt, \quad f \in \mathbb{R}.$$

If moreover $G \in L^1$, the **inverse** recovers g for almost all t : $g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df$. We write $g \longleftrightarrow G$ (a **Fourier pair**): no information is lost. Upper case denotes the transform.

Definition — Convolution

For $g, h \in L^1$, $u = g * h$ is $u(t) = \int_{-\infty}^{\infty} g(t-u) h(u) du$. It is a **weighted average** of one function by the other, centred at t .

Definition — Discrete-time Fourier transform

For a sequence $\{g_t\}_{t \in \Delta\mathbb{Z}} \in \ell^1$,

$$G(f) = \Delta \sum_{t \in \Delta\mathbb{Z}} g_t e^{-2\pi i t f}, \quad g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i t f} df.$$

The inverse is just a Fourier series with the domains swapped. Key feature: G is **periodic**, $G(f) = G(f + k/\Delta)$, so it is fully determined on one period of length $1/\Delta$.

Definition — Finite-data transform and the Dirichlet kernel

With n observations at times $T = \{0, \Delta, \dots, (n-1)\Delta\}$,

$$G_n(f) = \sum_{t \in T} g_t e^{-2\pi i t f}.$$

G_n is the *full* transform times a window indicator $d_t = \frac{1}{\Delta} \mathbf{1}_T(t)$, whose transform is the **Dirichlet kernel**

$$\mathcal{D}_T(f) = \sum_{t \in T} e^{-2\pi i t f} = \frac{\sin(n\pi f \Delta)}{\sin(\pi f \Delta)} e^{-\pi i (n-1) f \Delta}.$$

Hence $G_n = \mathcal{D}_T * G$: a finite window **blurs** the true transform.

Definition — Fourier frequencies

The n equally spaced frequencies (over one period) at which the FFT evaluates G_n :

$$f_k = \frac{k}{\Delta n}, \quad -\frac{n}{2} \leq k \leq \frac{n-1}{2} \quad \left(= \frac{k}{n} \text{ if } \Delta = 1 \right).$$

Key results & formulas

Theorem — Fourier inversion

If $g, G \in L^1$ (continuous time) or $\{g_t\} \in \ell^1$ (discrete time), the inverse transform recovers g . *Idea:* substitute the forward transform and use the orthogonality relation $\int e^{2\pi i f(t-s)} df$, which acts like a delta. This is what makes (g, G) a genuine pair and underlies every time $\leftrightarrow$ frequency duality.

Theorem — Convolution theorem

Multiplication in one domain = convolution in the other. For $g, h \in L^1$:

$$\begin{aligned} u = g * h &\longleftrightarrow U = G \cdot H, \\ v = g \cdot h &\longleftrightarrow V = G * H \quad (\text{if } G, H \in L^1). \end{aligned}$$

In discrete time both pairs hold too: $h * g \leftrightarrow H \cdot G$ and $h \cdot g \leftrightarrow H * G$. *Idea:* plug the convolution into the definition, swap the order of integration (Fubini, since $\|g\|_1 \|h\|_1 < \infty$), and recognise the phase factor. This is the central computational identity (filtering, blurring, etc.).

Theorem — Aliasing and the Nyquist frequency

Sampling *folds* all the continuous-frequency content onto one period:

$$G(f) = \sum_{k \in \mathbb{Z}} G(f + \frac{k}{\Delta}).$$

The **Nyquist frequency** is $1/(2\Delta)$ ($= \frac{1}{2}$ if $\Delta = 1$): content above it is **not** recoverable. *Idea:* write $g(t)$ via both inversion formulas at $t \in \Delta\mathbb{Z}$, split $\mathbb{R}$ into blocks of length $1/\Delta$, shift each ($e^{2\pi i t k/\Delta} = 1$ for these t), and match.

Proposition — Fast Fourier Transform (FFT)

Evaluating G_n at the n Fourier frequencies costs $O(n \log n)$ (vs. $O(n^2)$ naively) by **divide-and-conquer** (even/odd indices, $\log_2 n$ levels). This makes large-scale spectral computation feasible.

Basic properties (same form in continuous and discrete time):

$$\begin{aligned} \overline{g(t)} &\longleftrightarrow \overline{G(-f)}, & g(\alpha t) &\longleftrightarrow \frac{1}{|\alpha|} G(f/\alpha), \\ g(t + \tau) &\longleftrightarrow G(f) e^{2\pi i \tau f}, & \alpha g(t) + h(t) &\longleftrightarrow \alpha G(f) + H(f). \end{aligned}$$

(Conjugation reflects; stretching time compresses frequency; a time shift multiplies by a phase; the transform is linear.)

Intuition

Convolution = product: the Swiss-army rule

The rule “*convolution in one domain $\leftrightarrow$ product in the other*” is the whole point of Fourier: an expensive operation (convolution) becomes a plain multiplication. That is why we go to the frequency domain to filter, and why the blurring caused by a finite window is written as a convolution with a kernel.

Why aliasing hurts

A *high*-frequency sinusoid sampled too slowly looks *exactly* like a low-frequency one — the “wagon-wheel” effect in films. Frequencies f and $f + k/\Delta$ become indistinguishable: they are “aliases”. Cure: sample finely (small Δ , large Nyquist) or low-pass filter *before* sampling.

Finite window = blurred vision

We never observe the whole sequence, only n points. Mathematically, observing = multiplying by an indicator. By the convolution theorem this **convolves** the true transform with the Dirichlet kernel $\mathcal{D}_T$: peaks get smeared and energy “leaks” into neighbouring frequencies. The larger n , the more concentrated $\mathcal{D}_T$, the less blurring.

Discrete = continuous, but periodic and folded

In continuous time G lives on all of $\mathbb{R}$. As soon as we sample, G becomes *periodic* (period $1/\Delta$): knowing it on $[-\frac{1}{2\Delta}, \frac{1}{2\Delta}]$ suffices. Aliasing says exactly how the continuous spectrum folds back into that period.

Time shift

Shifting a signal by τ does not change its amplitude content, only its phase: $g(t+\tau) \leftrightarrow G(f)e^{2\pi i\tau f}$, so $|G|$ is unchanged. Intuition: moving a sound earlier or later in time does not change *which* frequencies it contains.

Key takeaways

Key takeaways — the week’s reflexes

- Be able to write the **three** transforms (continuous, discrete, finite-data) and their inverses, and to recognise a **Fourier pair** $g \leftrightarrow G$.
- **Core reflex:** convolution $\leftrightarrow$ product. Use it to relate G_n to G via $G_n = \mathcal{D}_T * G$, and for any filtering computation.
- Know the **4 basic properties** (conjugation, scaling, shift, linearity) — one-line proofs by a change of variable / inserting a phase.
- **Aliasing:** $G(f) = \sum_k G(f + k/\Delta)$; above the Nyquist frequency $1/(2\Delta)$ the information is lost. Sample fast enough.
- In discrete time everything is **periodic** with period $1/\Delta$: work on one period only.
- The **FFT** gives G_n at the Fourier frequencies in $O(n \log n)$.
- *Next stop:* this machinery defines the spectral density $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau}$ — the transform of the ACVS.